

AI and Ethics: Trust, Cooperation & Society

Benjamin Kuipers
University of Michigan
Computer Science & Engineering

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Huge Recent Progress in AI

- Deep neural networks
- Large Language Models: ChatGPT, et al.
- Amazing progress on particular tasks:
 - Game playing: Chess, Go, etc.
 - Protein folding
 - Materials science
 - Weather prediction
 - etc.
- Are we approaching the point where AI becomes better than (or as good as) the best humans at every human task? What then?

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Worries About Technology Over the Centuries

- Socrates vs the written word.
- Religious authorities vs the printing press.
- Luddites vs the Jacquard loom.
- Charlie Chaplin in “*Modern Times*”.
- Magic out of control in “*The Sorcerer’s Apprentice*”.
- “*Desk Set*” with Tracy and Hepburn.
- HAL in “*2001: A Space Odyssey*”.
- SkyNet in “*Terminator 2*”.
- . . . and many more . . .

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Worries About AI

- Will AI take all/most of the jobs, leaving most people in desperate straits?
- Will AI embody/enforce human bias against minorities?
- Will AI eliminate privacy, collecting vast data to track all of us, everywhere, all the time?
- Will AI take inhumane actions based on its own notion of rationality, harming individuals and perhaps humanity as a whole?

How can we think about these diverse concerns?

Can ethics help? What is ethics, anyway?

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Can ethics help? What is ethics, anyway?

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Existential Threats to Society

1. Climate Change
2. Infectious Diseases
3. Nuclear War
 -
 -
 -
- Super-Intelligent AI
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 -
 -

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Individuals Decide What To Do

- Individuals act
 - to make progress toward their goals
 - in the world as they understand it.
 - “Theory of Mind”
- “Rational” decision-making (Game Theory)
 - (inspired by recreational games)
 - formalizes this in terms of a utility function,
 - and selects actions to maximize utility.
- An important concept from Game Theory:
 - positive-, negative-, and zero-sum games.

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Individuals Belong To Societies

- Individuals thrive when their society prospers.
- Society prospers when interactions are **positive-sum**.
 - Positive-sum interactions are **cooperation**.
 - Cooperation involves **vulnerability** (to partners).
 - Vulnerability requires **trust** (of partners).
 - Trust pays off when the partners are **trustworthy**.
- Members of society are taught:
 - How to **be** trustworthy;
 - How to **signal** that **you** are trustworthy;
 - How to **recognize** that **others** are trustworthy.

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Society Needs Cooperation

- **Cooperation** is positive-sum interaction
 - among individual members of society.
- Cooperation requires **trust**.
 - *Trust is a psychological state comprising the intention to accept vulnerability based on positive expectations of the intentions or behavior of another.* [Rousseau, et al, 1998]
- Trust has value when partners are **trustworthy**.
- **Ethics** is how society teaches individuals how to be trustworthy.

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How Do We Observe Trustworthiness?

- **Theory of Mind** (one kind of commonsense knowledge)
 - *Agents select actions to approach their goals in the world as they understand it.*
 - Observing an agent's *action*, we infer its *goal*.
 - Knowing an agent's *goal*, we predict its *action*.
 - Knowing its *goal*, observing its *action*, we infer what it believes about its *world*.
- Learning an agent's goals in one situation
 - tells us persistent properties of that agent, such as values and character, that affect future decisions.
- Trustworthiness is a complex characteristic.
 - One can be trustworthy for one purpose, but not another.

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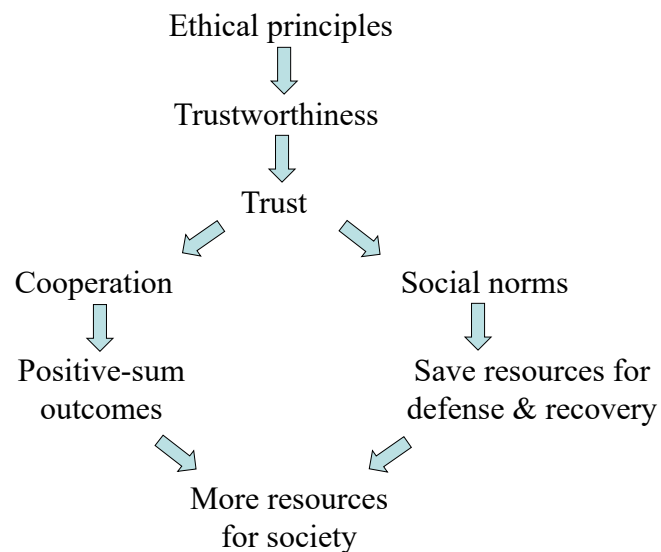
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Ethics Teaches Trustworthiness

- Ethics teaches society's members:
 - How to **be** trustworthy;
 - How to **signal** that **you** are trustworthy;
 - How to **recognize** that **others** are trustworthy.
- Evolutionary success of society depends on:
 - How well ethics encourages **positive-sum** outcomes.
 - a key concept from Game Theory
 - How well ethics is transmitted to new generations.
 - Important role for normative ethics.
 - How to describe and teach ethics.

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Trust is the Central Concept



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Human Evolution

- Cooperation allows groups (“societies”) working together to achieve much more than individuals.
 - Many animals have a useful *Theory of Mind*.
 - Predict what another individual is likely to do.
 - Humans evolved *joint intentionality* (400K ya)
 - Pairwise coordination toward the goal of a virtual “we”.
 - *Collective intentionality* evolved (100K ya)
 - The virtual “we” is the entire group. [Tomasello, 2019]
- Cooperation supports larger-scale societies. [Wright, 2000]
 - Cultural evolution accumulates knowledge, skills. [Henrich, 2016]
 - Individuals and societies able to cooperate thrived.
 - Others didn’t.

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Coping with Existential Threats

- Humanity faces existential threats:
 - Nuclear war
 - Global pandemics
 - Climate change
- Responding to these threats requires global **cooperation**
 - which requires **trust**, and **trustworthiness**.
- Can we do this?
 - Our existence may depend on our success.
 - “*Darwin could take us away!*”

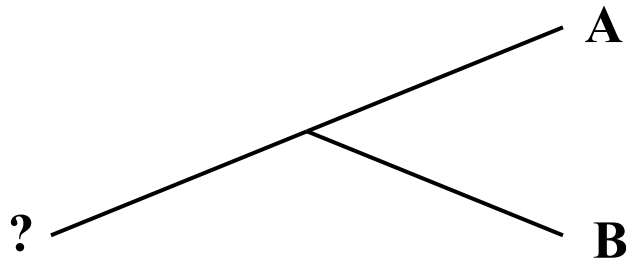
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The Big and Small Pictures

- That was the Big Picture:
 - How does ethics help our society (and us) succeed?
 - [Kuipers, Trust and cooperation,
Frontiers in Robotics and AI, 2022]
 - The existence of our society could be at stake.
- Now consider the Small Picture:
 - In a given situation, what should an individual do?
 - An AI is a decision-making individual in our society.
How can it be trustworthy? Can ethics help?

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A Choice Between A and B

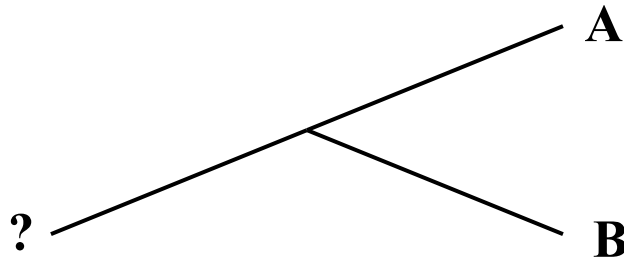


- What should we do?
 - Choose the **greater good**.
 - Choose the **lesser harm**.

if $u(\mathbf{A}) \geq u(\mathbf{B})$ then do(A) else do(B)

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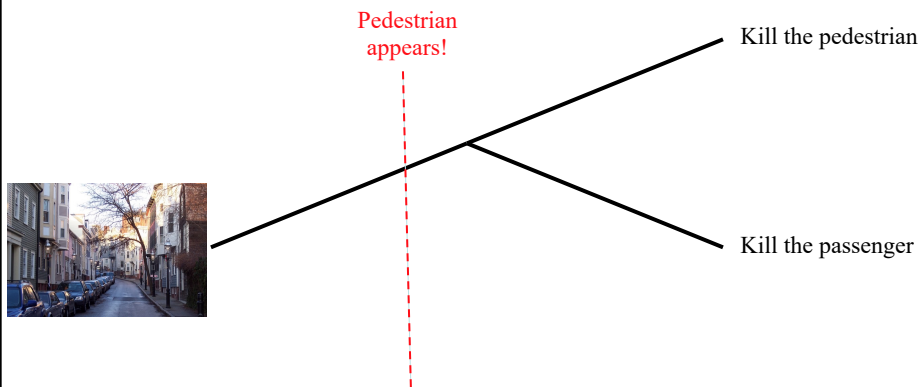
Tough Choices



- Sometimes **A** and **B** are both great, but different.
 - A tough choice between unknown futures.
- Sometimes **A** and **B** are both terrible.
 - A “Deadly Dilemma” can be heart-breaking.

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The Deadly Dilemma

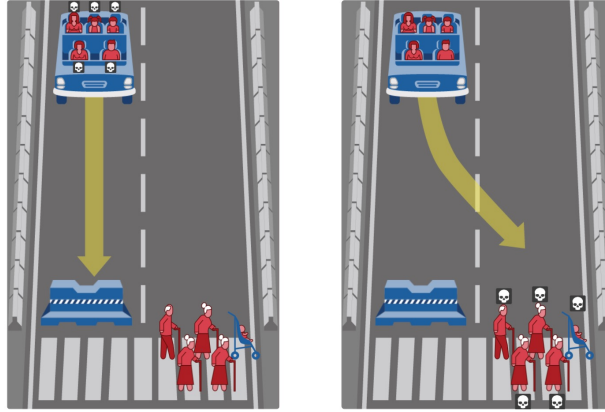


- This is sometimes presented as an ethical dilemma for autonomous vehicles (AVs).
 - A version of the “Trolley Problem”.

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A choice between two bad options



- The Moral Machine online survey.
 - Which action do you take?

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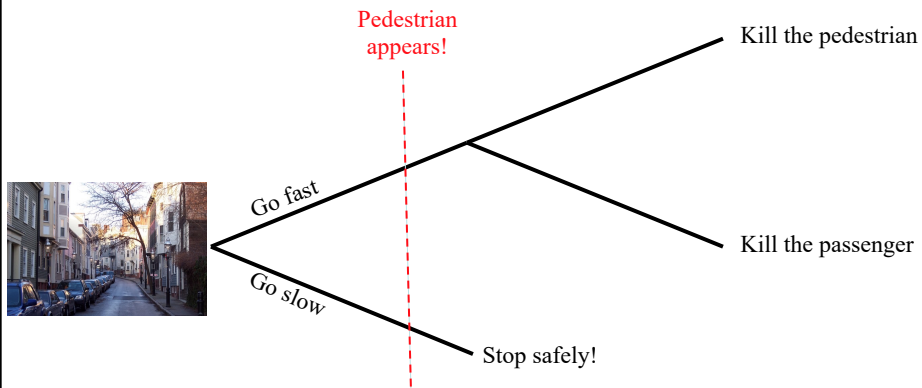
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Can We Avoid the Deadly Dilemma?

- Can the choice between harms be avoided?
 - Is there an “*upstream decision point*” where the right choice avoids both harms?
 - We teach this as part of the skill of driving.
 - If an upstream decision point exists, failure to recognize it and act then is considered *negligence*.
- The upstream decision point may not exist.
 - There are genuine moral dilemmas.
 - Different people may weigh harms differently.

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The Upstream Decision Point!

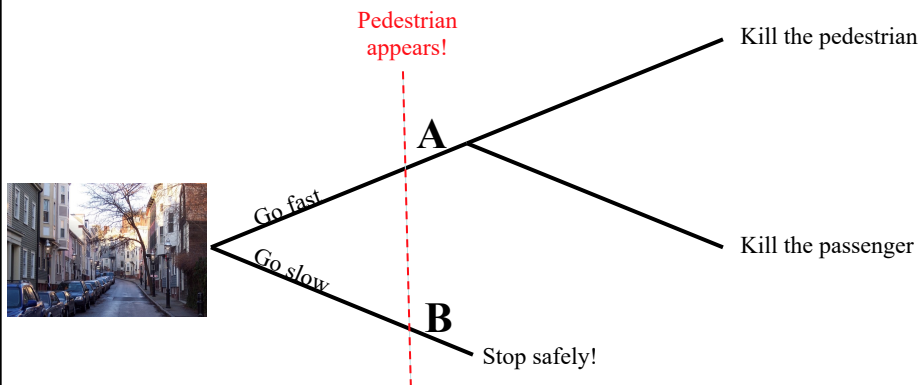


- The *actual* ethical choice occurred earlier.
- At a *previous* decision point, a different choice would have avoided the Deadly Dilemma entirely!

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The Upstream Decision Point!



- Both choices on the upper branch are terrible!

$$u(\mathbf{A}) \ll u(\mathbf{B})$$

$$\text{do}(\text{Go slow})$$

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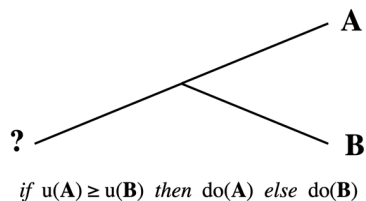
But You Can't Go Back in Time!

- The *teacher*, the *student* (and the *programmer*) can (and should) look forward in time!
 - Anticipate the Deadly Dilemma.
 - Learn from a Near Miss, or from someone else's terrible experience.
 - Search for an upstream decision point.
 - It often exists, but not always.
 - Learn to recognize that decision point, and what to do.
- Ethics is a skill, learned and practiced.
 - Aristotle said that ethics is a skill like carpentry.
 - “practical wisdom” = “*phronesis*” in Greek.
 - With that skill, tough choices are fewer. Life is better.

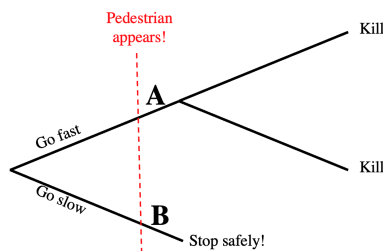
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Two Pieces of the Ethical Puzzle

- Do the most good / least harm [Utilitarianism]



- Learn the right actions at upstream decision points [Deontology]



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We Use Models of the World

- The world is infinitely complex.
 - To reason about it, we use simplified **models**.
- A model makes explicit a few aspects.
 - The *known unknowns*.
- Everything else is omitted as negligible.
 - The *unknown unknowns*.
- Watch out for neglecting important aspects.
 - For example, from the utility function.

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The Prisoner's Dilemma

- Two players work together for rewards*:
 - If you and your partner **Cooperate**, you each get a reward of 3 (dollars? gold bars?).
 - If you **Defect**, and your partner **Cooperates**, you get 5, and your partner gets 0.
 - If both **Defect**, both get 1.

	Cooperate	Defect
Cooperate	(3, 3)	(0, 5)
Defect	(5, 0)	(1, 1)

Utility is
gold bars.

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The Prisoner's Dilemma

	Cooperate	Defect
Cooperate	(3, 3)	(0, 5)
Defect	(5, 0)	(1, 1)

- Whatever your partner does, **Defect** is your best choice: $u(\text{Defect}) > u(\text{Cooperate})$
 - But your partner is making the same decision!
- The “*rational*” outcome is (**Defect, Defect**): (1, 1).
 - The *worst* outcome for society: 2.
 - And a bad outcome for you individually: 1.

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Why Such a Bad Outcome?

- The cooperative solution to PD is clearly better.
 - But it requires *trust*. And *trustworthiness*.
 - And those are not included in the model: $u(\mathbf{A})$.
 - When humans play these games, they consider alternate models and make better choices.
 - Except for majors in business and economics.
- When trust is omitted, utility maximization within this simple model exploits vulnerability.
 - Individuals learn not to trust.
 - Society loses the positive-sum benefits of cooperation.

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A Third Piece of the Ethical Puzzle

- What is being left out of the model?
 - when “rational” decisions lead to bad outcomes
- Often: the need for trust and trustworthiness
 - and their roles in cooperation.
- We need to ask what we need to trust.
 - How am I vulnerable (in a given context)?
 - How could my vulnerability be exploited?
 - What social norm would prevent that exploitation?
 - How should norm violators be sanctioned?
- With trust included, the outcome is better!

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Common Problems in AI Ethics

- Safe decisions by autonomous agents
 - Surveillance and privacy
 - Bias and fairness
 - Jobs and economic inequality
- These and other specific problems are where we should apply the Ethics, Trust, Cooperation (ETC) framework.

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Safe Decisions

- The problem is *situational awareness*, not *value alignment*.
 - If an AI's model of the situation is adequate, deciding on the best (or right) action should not be difficult.
 - But if significant hazards are invisible to it, it can unknowingly carry out harmful plans.
- How can the decisions of an AI be trustworthy?
 - We need to audit its *situational awareness*.
 - Are there important hazards that are invisible to it?
 - Tesla crashes into emergency vehicles stopped on the road.

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Surveillance and Privacy

- “Creepy omniscience” of websites.
 - “creepy” means “*sensing the potential for future harm*”
- Social networks and search engines sell your data.
 - Your data is valuable to advertisers and data brokers.
 - Lots of weak data creates strong personal models.
 - Maximize engagement to get more attention and data.
 - Stoking outrage increases engagement.
- You want to trust that your data is used responsibly and protect your privacy.
 - How can we regulate, and enforce regulations?

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Fairness and Bias

- Important decisions are made using data models.
 - Admissions, hiring, identifying suspects, bail, parole
- How can I trust that:
 - I am considered as an individual
 - not simply as a member of a category.
- Decision models often include bias.
 - Historical bias is embedded in training data.
 - Minorities are under-represented, decreasing accuracy.
 - Protected attributes are inferred from proxies.
 - Different formal definitions of *fairness*.

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Jobs and Economic Inequality

- Automation creates more wealth for society.
 - It also eliminates some jobs.
- People need meaningful work.
- There are jobs that are valuable for a good society.
 - e.g. “care” jobs: child care, elder care, health care, . . .
 - just like public schools, fire departments, defense, . . .
 - These are not profit centers. They are important.
 - They need subsidy.
- Can we use some of this wealth to create good jobs to support people in a good society?

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Which Jobs Should Be Created?

- People need work. Society needs resources.
 - Jobs are how society provides meaningful work.
- Can we create a market-based entrepreneurial process for creating new subsidized jobs?
 - Markets are powerful mechanisms.
 - If profit creation is the only utility measure on jobs, important jobs are overlooked.
- What do we need to trust in our society?
 - How can we create jobs to achieve that trust?

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Another Oversimplified Model With Destructive Consequences

*“The Social Responsibility of Business
Is to Increase its Profits”*

Milton Friedman, 1970.

- Very persuasive and influential.
- Deeply deceptive and irresponsible.

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What Have We Accomplished?

- We've put **trust** and **trustworthiness** at the center of our consideration of ethics and AI.
 - Ethics teaches us how to *be* trustworthy, and to *demonstrate* and *recognize* trustworthiness.
- When making plans and action decisions
 - Be sure our decision model includes what's important.
 - Maximize the good / minimize the harm.
 - Find upstream decision points to avoid dilemmas.

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Summary

- Trust, trustworthiness, and cooperation are central to society having the resources to thrive.
- Three pieces of the Ethics puzzle:
 - Choose to do the most good / least harm. [Utilitarianism]
 - Learn the upstream decision points. [Deontology]
 - You *must* use models. They *must* omit many things. Make sure the omitted things are *not* critical.
- Apply these principles in many specific contexts.
 - Surveillance & privacy, bias & fairness, jobs, safety, . . .

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Recommended Books

- Elinor Ostrom, **Governing the Commons**, 1990.
- Robert Wright, **Nonzero**. 2000.
- Bruce Schneier, **Liars & Outliers**, 2012.
- Joseph Henrich, **The Secret of Our Success**, 2016.
- Jay Van Bavel & Dominic Packer, **The Power of Us**, 2021.
- Daron Acemoglu & Simon Johnson, **Power and Progress**, 2023.
 - Plus many others!

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My Own Writings

- My web page: <https://web.eecs.umich.edu/~kuipers/>
- *AI and Society: Ethics, Trust, and Cooperation* (2023)
 - <https://web.eecs.umich.edu/~kuipers/research/pubs/Kuipers-cacm-23.html>
- *Trust and Cooperation* (2022).
 - <https://web.eecs.umich.edu/~kuipers/research/pubs/Kuipers-frai-22.html>
- Overview chapter on AI Ethics (2020)
 - <https://web.eecs.umich.edu/~kuipers/research/pubs/Kuipers-oheai-20.html>
- *How Can We Trust a Robot?* (CACM, 2018)
 - <https://web.eecs.umich.edu/~kuipers/research/pubs/Kuipers-cacm-18.html>
- *An Existing, Ecologically-Successful Genus of Collectively Intelligent Artificial Creatures* (2012)
 - <https://web.eecs.umich.edu/~kuipers/research/pubs/Kuipers-ci-12.html>
- All of my papers on AI and Ethics:
 - <https://web.eecs.umich.edu/~kuipers/research/ethics/papers.html>

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Thank You!